



Effects of cyclic hydrostatic pressure on the water absorption of polymer matrix solid buoyancy material

Wenjie Zhou¹ · Lei Chen² · Guoting Liu² · Fushun Liang² · Xiaoying Yan² · Peng Dong^{1,2}

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Abstract

From the perspective of service, the effects of cyclic hydrostatic pressure on the water absorption of polymer-matrix solid buoyancy material (PSBM) are critical for the submarine safety. However, the underlying mechanism has not been fully clarified to date. This work is focused on the performance aging and underlying mechanisms of PSBM under cyclic hydrostatic pressure. To comprehensively exhibit the actual service, a drying process was incorporated after each cycle during the hydrostatic pressure tests. The results show that the water absorption of PSBM increases with the number of uses, and long-term stress relaxation induces continuous internal stress redistribution, which exacerbates local stress concentration and ultimately triggers the initiation of microcracks. Stress relaxation is attributed to such performance degradation. Moreover, a model correlating the number of cycles to water absorption behavior was established, which enables the prediction of PSBM water absorption behavior during cyclic service. These findings are beneficial for improving the safety of submersibles and provide significant insights for the design of buoyancy materials in engineering applications.

Keywords Solid buoyancy material · Water absorption · Cyclic hydrostatic pressure · Lifespan prediction · Cyclic aging

1 Introduction

Polymer-matrix solid buoyancy material (PSBM) is a solid compound formed through physical and chemical reactions by filling inorganic materials such as glass microspheres into the resin matrix (Xu and Zhang 2016), which is necessary for the submarine by providing the buoyancy to achieve deep diving and surfacing. It has become the first choice buoyancy material in the deep-sea (Chenggui et al. 2017) field, because of its relatively low density, high strength, low water absorption, and low cost.

Water absorption (Li et al. 2009; Chen 2018) is one of the most critical properties for the PSBM. It not only affects the density and compressive strength but also determines the buoyancy it provides. Currently, the water absorption for almost all underwater equipment is required to be below 3% in initial tests, even some standards targeting rates less than

1%. At present, most of the research on the water absorption behavior of materials is to improve the composition and processing of materials. Jiyouping et al. (2023) proposed a vacuum-assisted isostatic pressing process. By fully accumulating the internal glass beads, the internal pore defects are reduced, thereby reducing the water absorption of the material. Long et al. (2016) used a variety of silane coupling agents to modify the surface of hollow glass microspheres, thereby reducing the gap between the matrix and the glass microspheres and the water absorption of the material. Liu et al. (2023a) used the vacuum composite casting process to increase the curing degree of the matrix by means of step-wise heating curing, thereby reducing the water absorption performance of the material. Unfortunately, people pay limited attention to the change of water absorption of PSBM after multiple cycles, whether in practice or scientific research.

It is important to clarify whether the water absorption of PSBM will increase after the repeated use of underwater equipment, and it is also important to know the number of cycles at which it will exceed the critical threshold. In fact, previous studies have shown that visible cracks (Cui et al. 2018) were found on the solid buoyancy material block of a deep-sea manned submersible after multiple uses. Kochetkov and Maksimov (1996) found that the water absorption of

✉ Peng Dong
dongpeng@tyut.edu.cn

¹ Department of Materials Science and Engineering, Taiyuan University of Technology, Taiyuan 030024, China

² Shanxi Fenxi Heavy Industry Co. Ltd., Taiyuan 030027, China

solid buoyancy materials continuously increases with time when placed in the marine environment, leading to a gradual decrease in compressive strength. Zhang and Zhang (2023) found that the tensile strength, elongation at break, compressive strength, tensile modulus and compressive modulus of solid buoyancy materials decreased significantly under cyclic hydrostatic pressure. In this case, it is reasonably deduced that the performance of the PSBM deteriorates with repeated use, resulting in a severe threat to the safety of the submersible and its occupants (Grosjean et al. 2009; Helal et al. 2019). It should be noted that current research and tests on the water absorption performance of solid buoyancy materials keep specimens immersed in water all the time, without considering that they may be removed from the water after a certain service and continue to serve after a period of time. The water absorption behavior of the material during this process could potentially differ from that when it is consistently submerged in water.

Therefore, cyclic hydrostatic pressure tests were carried out on PSBM specimens, with a drying procedure implemented after each loading cycle to simulate the practical condition where the material is retrieved from the water surface after service. The failure mechanism and water absorption behavior of PSBM during cyclic loading were investigated from the perspectives of microstructure evolution and mechanical property variation. On the basis of experimental data, a model was established to describe the correlation between cycle number and water absorption performance of PSBM, aiming to identify the critical transition point in property evolution. This critical point facilitates the prediction of the service life of PSBM under cyclic loading, thereby improving its reliability and stability.

2 Experimental details

2.1 Experimental materials and preparation

The main part of PSBM used is composed of hollow glass microspheres (HGM) and epoxy resin, with the remaining components being curing agent, toughening agent and fiber. A batch of the same PSBM samples was prepared for experimental analysis. The filler content is the ratio of the mass of the insulating glass microspheres to the mass of the entire composite material. Raw materials were obtained from the following suppliers: epoxy resin is bisphenol F epoxy resin (industrial grade), supplied by Nantong Xingchen Synthetic Materials Co., Ltd.; curing agent is methyl tetrahydrophthalic anhydride (Me THPA, industrial grade), supplied by Puyang Huicheng Electronic Materials Co., Ltd.; toughening agent is DMP-30; glass beads are hollow glass beads B; and fiber was supplied by Bengbu Glass Industry Design and Research Institute. The compressive strength of the tested

sample is 2.36 MPa, the density is 0.35 g/cm³, the diameter is 14.00 mm, and the height is 25.00 mm. The preparation process is shown in Fig. 1. The parameters of hollow glass microspheres (HGM) are shown in Table 1. The parameters of epoxy resin are shown in Table 2.

2.2 Cyclic loading test

First, utilize the lathe to remove the outer surface of the PSBM. Then weigh the material, and the initial mass was recorded as m_0 . After the infiltration treatment, and the material was placed in deionized water for immersion treatment, the treatment time was 30 s, and then the surface of the sample was immediately wiped dry with a towel, and the mass data were recorded as m_1 .

The loading equipment is a hydrostatic pressure simulation test device, as shown in Fig. 2a. Isostatic pressure is used for loading. In accordance with Miner's linear fatigue damage accumulation rule and continuum damage mechanics, the ultimate compressive loading test adopted herein is regarded as an accelerated aging approach, which can accelerate damage evolution and reveal the underlying aging failure mechanism under limited cycles. The PSBM specimen was loaded into the permeable fixture and placed into the external water pressure experimental autoclave. The applied pressure was set to 2.36 MPa, and the holding time was 10 h. Figure 2b shows the cyclic loading curve for three cycles.

According to this procedure, 30 cyclic loading tests were performed. Upon completion of pressurization, excess moisture on the surface of the PSBM specimen was immediately wiped off with a towel, followed by weighing, and the measured mass was recorded as m_2 . Next, the water absorption of the specimen was calculated using Eq. (1). Subsequently, the specimen was placed in an incubator and held at 60 °C for 5 h to ensure complete evaporation of penetrated water molecules. The above procedures were repeated to complete each full cycle. A pure thermal cycling control test without hydrostatic pressure was supplemented, in which specimens only underwent cyclic drying at 60 °C over 5 h. As shown in Fig. 2c, analysis of the control group data shows that the compressive strength remains nearly unchanged under pure thermal cycling. This result demonstrates that thermal aging can be distinguished from resin post-curing effects. It further verifies that cyclic hydrostatic loading and the accompanying structural damage dominate the mechanical degradation of the material.

$$m = \frac{m_2 - m_1}{m_0} \times 100\% \quad (1)$$

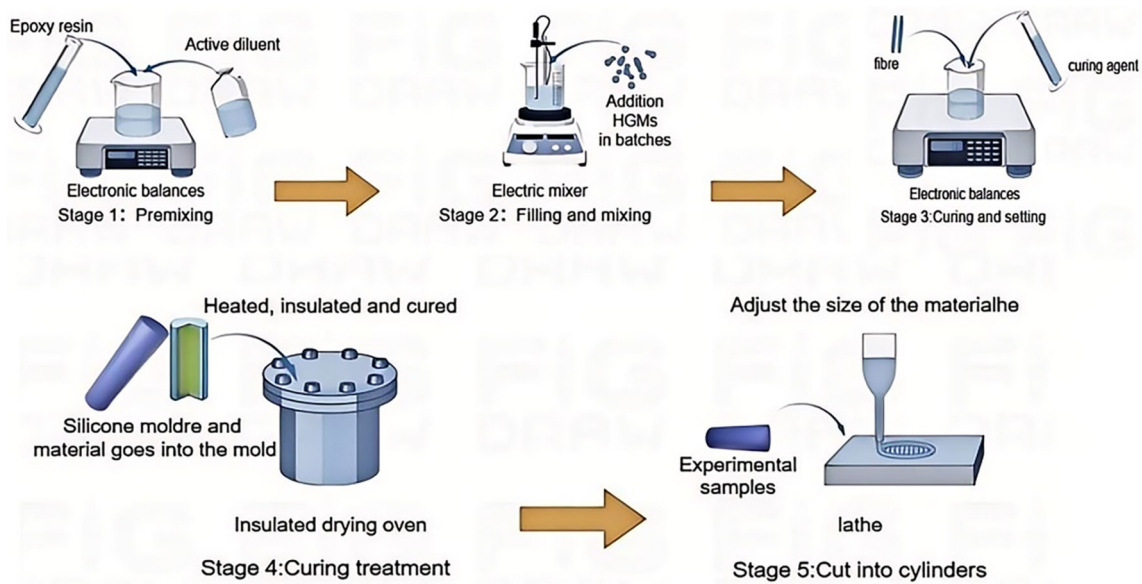


Fig. 1 Fabrication flow chart of the polymer matrix solid buoyancy material composites

Table 1 Calculation table of various parameters of the hollow glass microspheres (HGM)

HGM type	Extrusion strength (MPa)	Typical density (g/cm ³)	Appearance	Particle size distribution (μm)
B	13	0.25	White powder	<90

Table 2 Calculation table of various parameters of the epoxy

Resin type	Compressive strength (MPa)	Tensile strength (MPa)	Viscosity (Pa s)	Density (g/cm ³)	Appearance
Bisphenol F epoxy resin	11.3	13.7	3.8	1.12	White liquid

3 Results and discussion

3.1 Cyclic water absorption behavior

Figure 3 illustrates that the water absorption of PSBM generally increases with the number of cycles. Experimental data indicate that the water absorption evolution can be divided into three distinct stages. In the first stage, the water absorption presents a nonlinear variation. At the early cyclic stage, the water absorption of PSBM fluctuates obviously with increasing hydrostatic pressure cycles. As the cycle number further increases, the water absorption enters the second stage, in which the variation gradually decreases and tends to converge. After a period of stable water absorption, the value rises sharply, corresponding to the third evolutionary stage. Unlike the second stage, the third stage is characterized by a rapid increase in water absorption. Significant differences in surface morphology can be observed among the three stages, as displayed in Fig. 3. In the first stage (Fig. 3a), no obvious pores or defects appear on the material surface. In the second

stage (Fig. 3b, c), tiny cracks and pores gradually generate on the surface. In the third stage (Fig. 3d), prominent surface cracks emerge. The compressive strength evolution of the composite under cyclic loading is summarized in Fig. 3f. The curves demonstrate that the compressive strength generally decreases with the increase in water absorption and presents an aggravated degradation trend as the number of cycles increases.

There are two main reasons for the difference in water absorption behavior of materials (Gomez et al. 2021). One is the water absorption of the material itself, and the other is the pores in the composite material. The main matrix of the solid buoyancy material is a mixture of cured epoxy resin and glass beads. As a polymer organic material, although it has a certain water absorption, the incorporation of beads will produce some small pores and the cavity after the break of the beads. Therefore, the main contributor to water absorption of the solid buoyancy material is the pores (Hardiman et al. 2016).

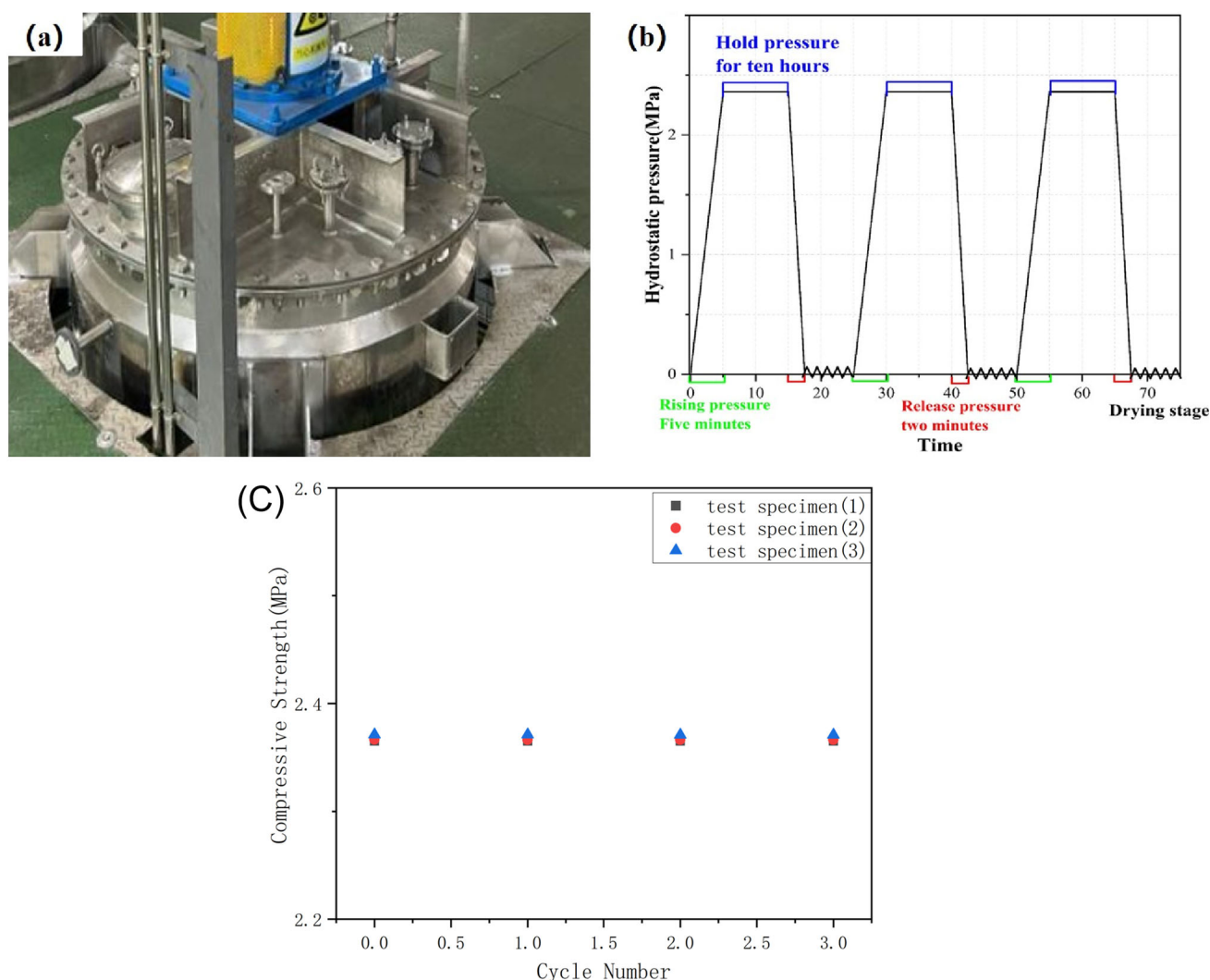


Fig. 2 **a** External water pressure experimental kettle. **b** Cyclic loading curve (three cycles). **c** Compressive strength versus cycle number for pure thermal cycling control tests without hydrostatic pressure

Based on these observations, it can be hypothesized that the underlying reasons for the observed changes are as follows. In the early loading stage, cyclic loads induce instantaneous elastic deformation of the material; the temporary opening and closure of pores and microcavities interfere with the transport and diffusion of water molecules, which consequently causes a fluctuation in water absorption rate. In the second stage, the water absorption capacity of the material gradually stabilizes because the channels for water molecules to enter the PSBM remain unobstructed. As the experiment progresses, the internal water absorption capacity of the material reaches saturation, resulting in a state of equilibrium. In the third stage, the continued increase in water absorption capacity is caused by the expansion of the internal channels, which occurs as the experiment advances, leading

to structural damage within the material and the formation of microcracks on the surface.

3.2 Stress relaxation test

Polymer materials have macromolecular chain structure and unique thermal motion, which determine that they have different physical properties from low molecular materials. The most important feature of the mechanical behavior of polymer materials is that they possess high elasticity and viscoelasticity. Under the action of external force and energy, the mechanical behavior of polymer materials is more strongly affected by factors such as temperature and time than that of metal materials, and their mechanical properties change greatly. The deformation of the polymer is achieved by adjusting the internal molecular conformation (Keddie et al.

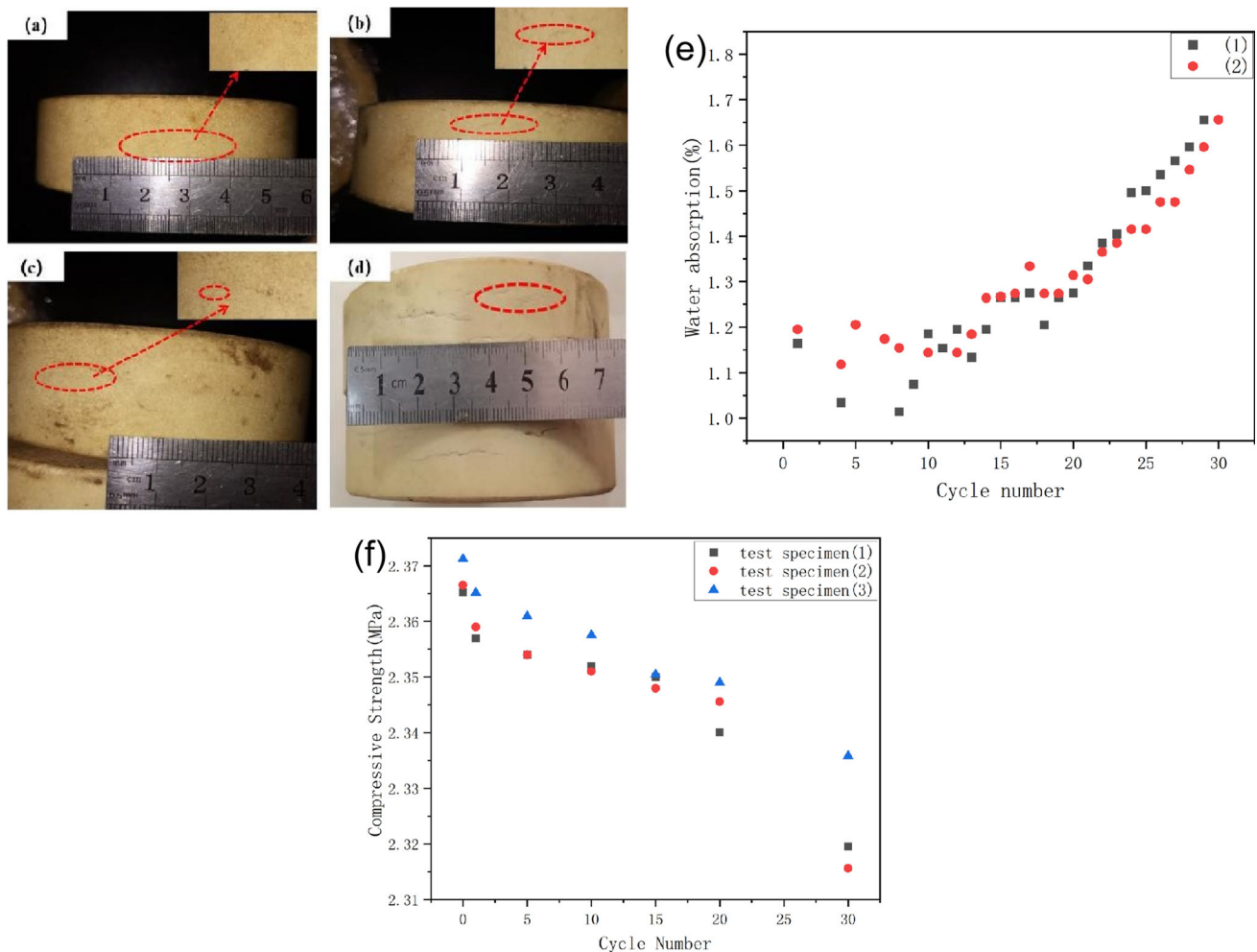


Fig. 3 **a** The surface of the first stage of the experimental cycle. **b** The surface of the second stage of the experimental cycle. **c** Different surfaces in the second stage of the experimental cycle. **d** The surface of the third stage of the experimental cycle. **e** Water absorption of specimens

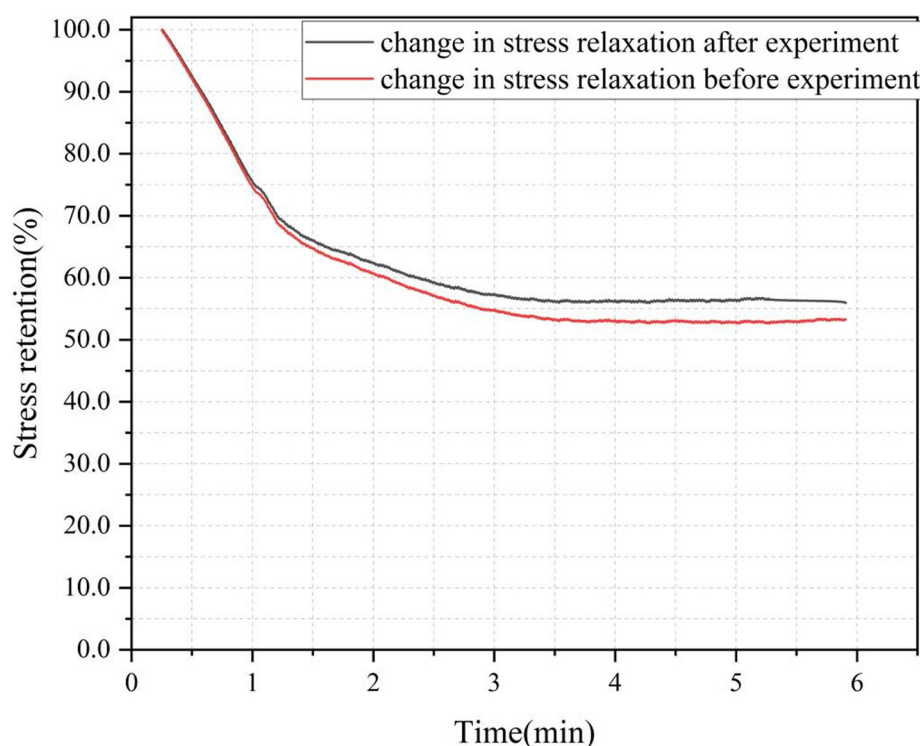
under different cycle numbers. Each data point represents the average value of three parallel tests ($n = 3$). **f** Changes in compressive strength of specimens with varying cycle numbers. Compressive strength vs. cycle number. $n = 3$ per data point

1994). Stress is always involved during the forming process. In the process of curing into a product, residual internal stress is inevitably trapped inside the composites during curing. Such residual stress will slowly relax during the storage and use of the product, resulting in warpage, deformation and even stress cracking of the product. Stress that is not fully relieved during curing will be more or less frozen within the product. Since PSBM is fabricated under certain pressure conditions, regions with relatively high internal stress are inevitably formed (Liu et al. 2023b). We further investigated the cracking mechanism. The analysis demonstrates that cyclic loading accumulates fatigue damage continuously, coupled with stress relaxation under long-term service environments. Their synergistic effect facilitates the initiation

and propagation of internal damage in PSBM. Corresponding experiments were conducted to confirm the presence of stress relaxation inside the material.

The PSBM is made into a stress relaxation sample according to the test requirements, and the stress relaxation test of the sample is carried out before and after the hydrostatic pressure cyclic load test in compliance with GB/T 9871–2008. The stress relaxation curve of the material within 360 s after stretching was determined. The stress relaxation condition is 0.8 times the elongation at break. In the initial stage, the stress decay rate is relatively fast, and the stress decay rate gradually decreases with time, and finally tends to the equilibrium value. This indicates that the viscoelastic performance of the material is essentially determined by the viscoelastic performance of the epoxy resin.

Fig. 4 Stress relaxation curves before and after cyclic testing of polymer matrix solid buoyancy material



The stress relaxation curves of the PSBM before and after the cyclic loading test are similar. The curves show that the PSBM has a large amount of stress relaxation. As shown in Fig. 4, the relaxation amount of the sample after the cyclic loading test is close to 46.8%, and the relaxation amount of the sample before the cyclic loading test is 43.9%. Contrary to the above numerical comparison, after the external pressure cycle aging experiment, the relaxation of the PSBM is reduced, which indicates that the long-term external pressure cycle service will lead to an increase in the stress relaxation of the PSBM, which not only weakens its viscoelasticity, but also reduces compressive strength and water absorption. In the process of recycling, the molecular chain of the PSBM will be elongated, and the internal stress will gradually decrease with time.

The compressive strengths of pristine PSBM specimens and samples subjected to 20 cyclic loading cycles were tested. The results show that the compressive strength of the material exhibits obvious degradation after 20 cycles, which is consistent with the trend presented in Fig. 3f: significant changes in mechanical properties occur around 20 cycles. Given that abrupt variations in compressive strength and water absorption are observed at 20 cycles, this characteristic node was selected to conduct stress relaxation tests. The test results verify that stress relaxation originates from the slow motion of polymer molecular chains. Comprehensive analysis demonstrates that cumulative fatigue damage continuously develops with increasing cycles. Combined with stress redistribution induced by stress relaxation, the synergistic effect of these

two factors continuously accelerates the performance degradation of the material.

3.3 Micromorphologies of buoyant materials

The fracture morphology of PSBM was characterized by optical microscopy, as shown in Fig. 5. Scanning electron microscopy (SEM) was further adopted to observe microstructural changes of specimens before and after cyclic loading. The SEM results revealed that the hollow glass beads inside the material exhibit no evident damage throughout the cyclic test. It is concluded that the variation in material performance during cyclic loading mainly stems from matrix property deterioration, which induces material aging and mechanical degradation. The results show that the overall mechanical behavior of buoyant materials is strongly affected by the properties of the matrix (d'Almeida and Monteiro 1998).

In Fig. 6a, small cracks and pores inevitably exist on the surface and interior of the original PSBM specimen. Therefore, under cyclic external hydrostatic pressure, water molecules penetrate into the interior of the material through surface microcracks and pores. As the number of cycles increases, water molecules gradually diffuse into the internal high-stress regions. Under the coupling effect of water infiltration and cyclic loading, stress relaxation first occurs in the high-stress regions of PSBM, eventually resulting in the initiation and propagation of internal cracks.

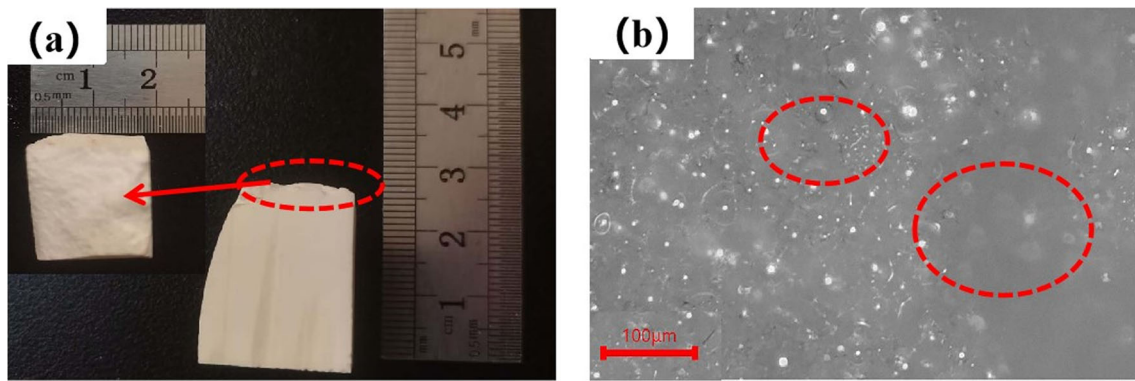


Fig. 5 **a** The fracture surface. **b** The fracture surface under the light microscope

A comparison of the micrographs before loading and after the 10th hydrostatic pressure cycle in Fig. 6a and b indicates no obvious variation of the overall internal pore structure of the specimens. As shown in Fig. 7a, when the number of cycles reaches 15, the interfacial voids between hollow glass microspheres and the resin matrix expand significantly. This provides additional penetration pathways for water molecules, macroscopically leading to a gradual increase in water absorption. After 30 cyclic loadings (Fig. 7b), SEM observations show evident interfacial debonding and delamination at the microsphere-matrix interface, which destroys the integrity of interfacial bonding.

Further analysis of material property variations indicates that such evolutions are not induced by changes in the internal composition of the material. Meanwhile, EDS analysis was conducted to characterize the microstructural evolution within the PSBM matrix. As shown in Fig. 6e, the primary constituents of PSBM after cyclic tests remain cured epoxy resin and hollow glass microspheres, and no new phases are generated. The dominant factor driving property evolution during cyclic loading is molecular chain relaxation, which results in the performance degradation of PSBM. Therefore, the chemical composition of PSBM remains unchanged under cyclic external hydrostatic pressure. The water absorption behavior corresponds to the simple inward diffusion of water molecules. Water penetration does not trigger irreversible physical or chemical reactions inside the PSBM.

In summary, under cyclic external hydrostatic pressure loading, the dominant factor limiting the service life of PSBM is internal stress relaxation. Stress relaxation preferentially takes place in high-stress regions, triggering the initiation of microcracks. As the service duration extends, these microcracks propagate from the interior outward and eventually develop into severe macroscopic cracking. The crack propagation process involves matrix cracking, interfacial debonding between the matrix and hollow glass beads, water molecule diffusion along internal channels, fracture of glass beads, and internal and edge delamination of the

composite. These damage modes interact synergistically and continuously evolve in the form of a damage field, constituting a progressive damage accumulation process. Therefore, it is concluded that the failure of PSBM under cyclic loading originates from the synergistic effect of internal stress relaxation and cumulative cyclic fatigue damage.

3.4 Model and verification

Poor dimensional stability is an inherent disadvantage of polymer materials. Stress relaxation often occurs under normal-temperature conditions, which leads to the failure of material structure. This factor must be considered in engineering design. Stress relaxation refers to the phenomenon that the internal stress of the material or component decreases with time. The phenomenon of stress relaxation is widespread, especially as a key part of the structural system. The phenomenon of stress relaxation has a serious impact on its working performance and bearing capacity. It will not only reduce the safety and service life of the material structure, but also limit its application in the industrial field. Therefore, this factor must be taken into account. The stress relaxation phenomenon will lead to the change of material structure performance. Because the internal viscous strain (viscoplastic strain) component increases with time, the rebound strain component gradually decreases with time, resulting in the phenomenon that the deformation restoring force (rebound stress) gradually decreases with time, and this process affects the reliability of its long-term service. On the premise of not affecting the basic characteristics of materials, it is of great significance to study the stress relaxation characteristics of solid buoyancy materials and establish the stress relaxation model for understanding their physical laws (Romanyk et al. 2014; Papanicolaou et al. 2008; Koyanagi et al. 2004; Guedes 2006).

The essence of solid buoyancy materials is polymers, and the process of polymer reaching equilibrium through physical aging is essentially the relaxation process of glassy

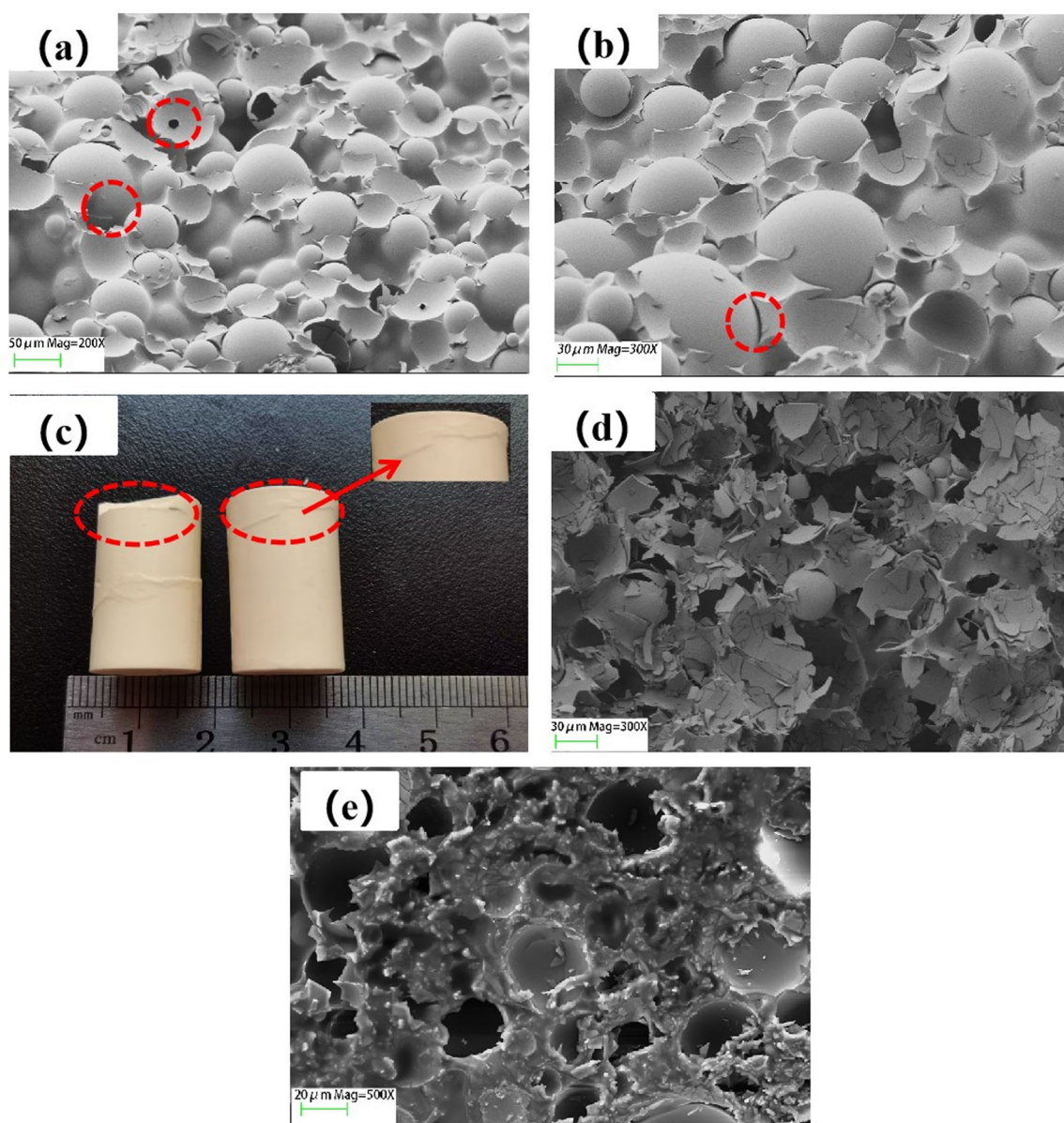


Fig. 6 SEM images of PSBM. **a** SEM before cyclic loading (Mag = ×200, scale bar = 50 μm). **b** SEM after cyclic loading (Mag = ×300, scale bar = 30 μm). **c** Samples after compressive strength test

before cyclic loading. **d** SEM after compressive strength test before cyclic loading (Mag = ×300, scale bar = 30 μm). **e** EDS after cyclic loading (Mag = ×500, scale bar = 20 μm)

molecular structure (Guedes 2008; Mano et al. 2005). Since the relaxation rate of the 'frozen' molecular chain is slow, the physical aging behavior has a long-term range of characteristics. It has been proved that the movement of molecular chain segments plays a decisive role in the structural relaxation of polymers. The chain segments in the high-energy conformation region of the polymer are transformed into a low-energy state through local conformational changes, which can be regarded as the essence of structural relaxation during aging (Cowie and Arrighi 2014; Wang 2006). Under the action of

large stress, the barrier of chain segment motion of glassy polymer is relatively reduced, thus shortening the relaxation time of polymer chain segment motion along the direction of external force, so that the frozen chain segment of glassy polymer can cross the barrier and move to a near-high elastic state. Experiments show that there is a relationship between relaxation time and stress (Yang 1999; Moynihan et al. 1976; Di Lisio et al. 2024).

$$\tau = \tau_0 \exp\left(\frac{E - \alpha\sigma}{RT}\right) \quad (2)$$

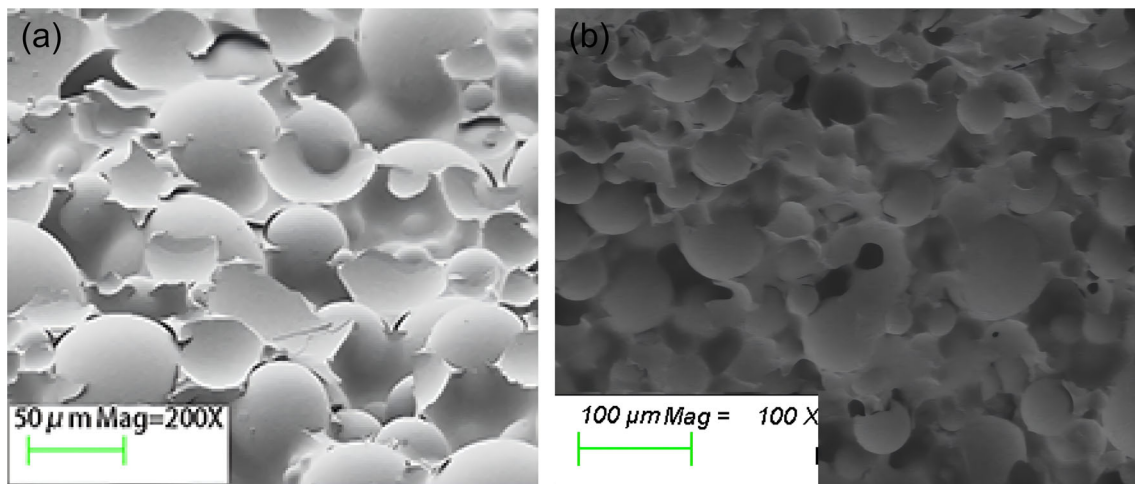


Fig. 7 SEM images of PSBM. **a** 15 SEM after cyclic loading (Mag = ×200, scale bar = 50 μm). **b** 30 SEM after cyclic loading (Mag = ×200, scale bar = 50 μm)

where τ_0 denotes the reference relaxation time; E is the activation energy of matrix resin (unit: J/mol, standard molar energy unit); σ denotes the cyclic equivalent hydrostatic pressure (Pa); α is the stress weakening coefficient, which depends on the material microstructure and is calibrated via stress relaxation tests; R denotes the universal gas constant J/(mol K); T denotes the thermodynamic temperature (K).

This formula infers that when the external stress of the material changes, the stress relaxation time of the material will also change. Therefore, as the external water–pressure cyclic experiment proceeds, water penetrates into the material, and the bearable pressure of the material gradually decreases, which continuously shortens the relaxation time, inducing stress relaxation of the material and generating internal cracks, such that the material performance fails to meet service requirements.

In the process of normal use and aging of solid buoyancy materials, it is considered that the pressure borne by the materials is within the scope of use, so the internal glass beads are not broken. Two competing processes, namely strength enhancement and damage degradation, occur during service (Moynihan et al. 1976). For thermosetting composites exposed to the environment under no load, assuming that the enhancement process and damage are independent of each other, the relationship between the median aging life (Bogdanov et al. 2023) and the residual strength of the composites can be determined by the following formula. Static corrosion-free aging model:

$$S = S_0 + \eta(1 - e^{-\lambda t}) - \beta \ln(1 + \theta t) \quad (3)$$

where η is the material parameter, reflecting the degree of curing of the material. Matrix post-curing strengthening coefficient (corresponding to the short-term strength

improvement induced by continuous resin curing); λ is the material and external environmental parameters, reflecting the strengthening rate characteristics; β reflects the corrosion parameter of external environment, which is governed by the combined effect of chemical corrosion parameter θ_C and water-induced physical damage coefficient θ_W ; S is the strength of the material after aging t time; S_0 is the initial strength of the material. The parameters η and β are only related to the material properties and can be determined by a series of accelerated aging tests. They can also be used to determine the λ and θ in the equation and extrapolate to the natural environment. $\ln(1 + \theta t)$ this term denotes the cumulative damage rate per unit time, which is governed by the stress relaxation rate of the material.

By establishing the equation of the number of cycles and the relaxation time, we combine (2) and (3) to further express the service life of the PSBM. The total duration of a single loading cycle t is fixed, consisting of pressure ramping, a 10-h pressure holding stage, immersion treatment, and a 5-h drying stage. The total service time has a definite correlation with the number of loading cycles.

$$t = Nt_c \quad (3-1)$$

Consistent with experimental observations: as the cycle number N increases, cumulative internal damage leads to an elevation in the effective bearing stress of the material. Meanwhile, the residual strength of the material is dominated by the degree of viscoelastic relaxation. Substitute Eq. (3–1) into Eq. (3) to accomplish variable transformation from time t to cycle number N . We obtain the relationship between the relaxation time and the number of cycles as follows:

$$\tau = \tau_0 \exp \left[\frac{E}{RT} - \frac{\alpha}{RT} S_0 - \frac{\alpha}{RT} \eta(1 - e^{-\lambda t}) + \frac{\alpha}{RT} \beta \ln(1 + \theta t) \right] \quad (4)$$

To simplify the calculation order: E , R , T , α , S_0 are inherent material-environment parameters and remain constant under identical experimental conditions. All constant terms are combined item by item to simplify the calculation procedure.

$$k = \frac{E}{RT} - \frac{\alpha}{RT} S_0 \quad (5)$$

$$a_1 = \frac{\alpha}{RT} \eta \quad (6)$$

$$a_2 = \frac{\alpha}{RT} \beta \quad (7)$$

Because the E , R , T , α , S_0 in the material are constants, they will be replaced by constants k , a_1 , a_2 . Simplified formula:

$$\tau = \tau_0 \exp[k - a_1(1 - e^{-\lambda t}) + a_2 \ln(1 + \theta t)] \quad (8)$$

In the process of this experiment, we can get the relationship between stress and cycle times. Using this, the relationship between the number of service can be obtained. θ_W refers to the water-induced physical damage coefficient. No chemical corrosion of resin is observed in the present material. Nevertheless, the penetration of water molecules into pores and matrix plasticization under wet-dry cycles are non-negligible, and thus θ_W cannot be set to zero. Experimental results indicate that interconnected microcracks initiate within the matrix once the relaxation time reaches the critical value. This critical condition is adopted for model simplification.

$$t = -\lambda \ln\left(1 + \frac{S}{\eta} - \frac{E}{\alpha \eta}\right) \quad (9)$$

Based on Fick's water diffusion theory and polymer plasticization theory, water absorption of the matrix is divided into two components: bulk water absorption and pore water absorption: The water absorption of the material is the water absorption of the matrix and the water absorption of the pores, and the relationship between the water absorption rate and the strength of the material matrix:

$$D_\sigma = D_0 \exp\left(\frac{\alpha}{E} k w_m\right) \quad (10)$$

where: D_0 is the initial diffusion coefficient; D_σ is the diffusion coefficient under stress; α is a constant, the value range is 6–10, E is the compressive strength of the matrix material, and w_m is the water absorption of the matrix. With the increase of the number of times, the force itself remains unchanged, but the force it can withstand gradually decreases, resulting in the appearance of porosity, so that the water absorption rate of the material continues to increase. Due to the plasticizing effect of the matrix, water penetration

leads to a decrease in the glass transition temperature of the polymer matrix:

$$E = E_0 \exp(-k w_{\text{total}}) \quad (11)$$

If we combine (9) with (11), we get:

$$t = -\lambda \ln\left(1 + \frac{S}{\eta} - \frac{E_0 \exp(-k w_{\text{total}})}{\alpha \eta}\right) \quad (12)$$

Since the water absorption of the PSBM in the early stage was fluctuating and could not be fitted, we fitted the 8th to 20th groups of the PSBM as shown in Fig. 4. Figure 4 shows that the intermediate materials of 1% and 1.25% are in the range of normal use, but the water absorption of the PSBM increases faster after the 20th group.

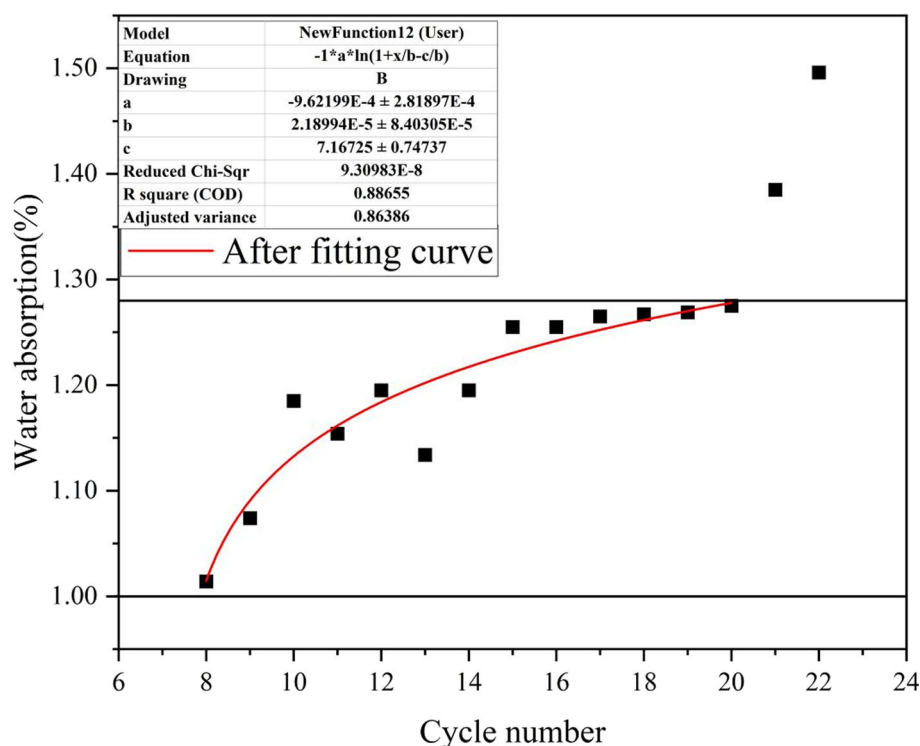
According to the simplified formula (12), the cycle life of the PSBM can be improved by increasing the curing degree of the PSBM and adjusting the structure. The number of cycles of the PSBM can also be inferred by this formula. This formula is fitted with the experimental results. As shown in Fig. 8, it is found that the fitting is consistent. When the maximum water absorption of the material reaches a horizontal state, the performance of the PSBM will reach a peak.

4 Conclusions

In this work, the cyclic aging process of PSBM was studied experimentally. The relationship between the number of cycles and the water absorption rate of the same PSBM under cyclic conditions was obtained. Then, the variation in PSBM service life was analyzed based on the change of water absorption in the external water pressure cycle test. The life prediction model was established using the relationship between the number of cycles and the water absorption, and the reasons for the change of the life of the material were analyzed by the change of the internal stress of the PSBM. Finally, from the microscopic level of scanning electron microscopy, energy spectrum analysis and mechanical properties analysis, it is concluded that the failure mechanism of PSBM is internal stress relaxation. The main conclusions were as follows:

1. The water absorption behavior of PSBM presents two distinct evolution modes. The water absorption in the early stage conforms to classical Fickian diffusion. As cyclic loading induces microcracks and structural damage, the water absorption process gradually transforms into damage-coupled non-Fickian diffusion. On this basis, the authors proposed an empirical formula to predict the water absorption of cycle times and water absorption. According to the curve of the simulation formula, it can

Fig. 8 The experimental data were fitted using Formula 12. Each data point represents the mean of three independent measurements ($n = 3$)



be determined that during the cycle of the PSBM, when the water absorption performance of the PSBM reaches equilibrium for a period of time, the performance of the PSBM will decrease seriously.

- After the cycle process, the water-absorption variation of PSBM is as follows: in the early stage of the material cycle, there is a fluctuation. At the initial loading stage, cyclic loads induce transient elastic deformation of the material. The repeated temporary opening and closing of pores and micro-gaps disrupt the diffusion of water molecules, resulting in unstable water absorption. After numerous cycles, the interfacial voids between microspheres and matrix expand considerably, and elastic deformation no longer hinders water penetration, so the material gradually reaches water saturation. With the increase in the number of cycles, further enlargement of interfacial gaps causes massive detachment of microspheres, which continuously raises the water uptake capacity and ultimately degrades the comprehensive performance of PSBM.
- Through SEM, spectral analysis and stress relaxation experiments, it is found that at the micro level, in the process of cyclic aging of materials, the main cause of PSBM failure is stress relaxation. Microcrack initiation is not solely caused by stress relaxation. The combined effect of cumulative fatigue damage under cyclic loading and stress redistribution induced by long-term stress relaxation leads to local stress concentration and finally triggers microcrack nucleation.

Due to the application of experimental equipment and methods, this study has some limitations and shortcomings. This paper does not consider the change of water absorption rate for a long time and the influence of seawater salinity on water absorption. In the future, this paper will continue to study the aging characteristics of solid buoyancy materials in the marine environment. Considering various environmental factors, the coupling effect of seawater salinity and long time on water absorption will be studied.

Author contributions Wenjie Zhou Writing—review & editing, Investigation, Formal analysis. Peng Dong Writing—review & editing, Supervision, Methodology Visualization, Formal analysis. Fushun Liang, Xiaoying Yan Writing—review & editing, Validation. Lei Chen, Guoting Liu, Writing—review & editing, Supervision. Peng Dong Writing—review & editing, Funding acquisition, Conceptualization, Resources.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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